Sediment Core Analysis

In [1]:
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from IPython.display import Image
from IPython.display import HTML
```

There is no safe level of lead. All exposure is bad. The next text is direct from the EPA websie https://www.epa.gov/lead/what-are-some-health-effects-lead

Lead is known to cause a range of health effects, from behavioral problems and learning disabilities, to seizures and death. Children under six years of age are most at risk from exposure to lead-based paint because they crawl on the floor and they put their hands and other items which can have lead-based paint dust on them into their mouths. Because their bodies are still growing, children tend to absorb more lead than adults.

Children exposed to lead can suffer from:

- Lowered IQ
- Damage to the brain and nervous system
- Learning and behavioral difficulties
- Slowed growth
- Hearing problems
- Headaches

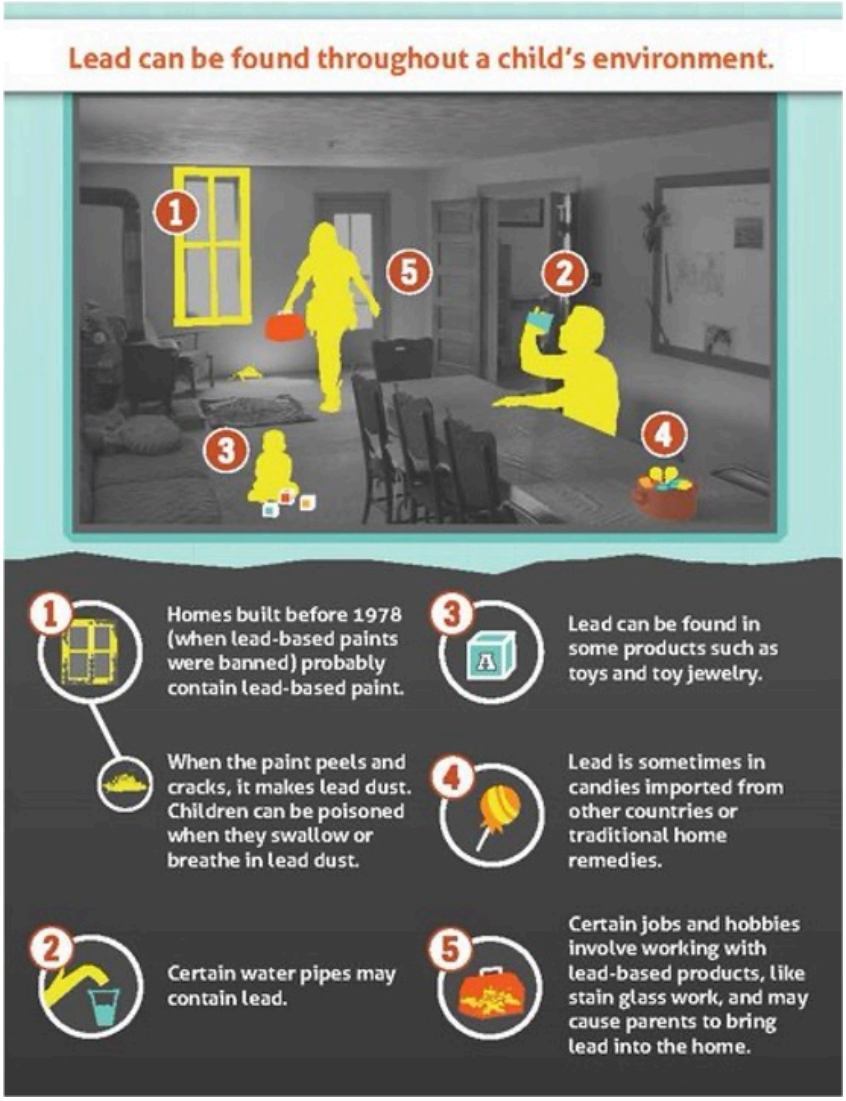
Sources of lead are from https://doh.wa.gov/community-and-environment/contaminants/lead/common-sources-lead-poisoning

- Lead-Based Paint
- Contaminated Soil
- Children's Toys and Jewelry
- Drinking Water
- Workplace and Hobby Hazards
- Traditional Home Remedies and Cosmetics
- Lead Glazed Ceramic Ware, Pottery and Leaded Crystal
- Imported Candy
- Mini Blinds
- Historically also leaded gasoline which contaminated soil (was a major source)

Schematic of Lead in a household from CDC

In [8]:
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Image(filename='Lead_in_house_cdc.jpg',width=600)
```

Out[8]:



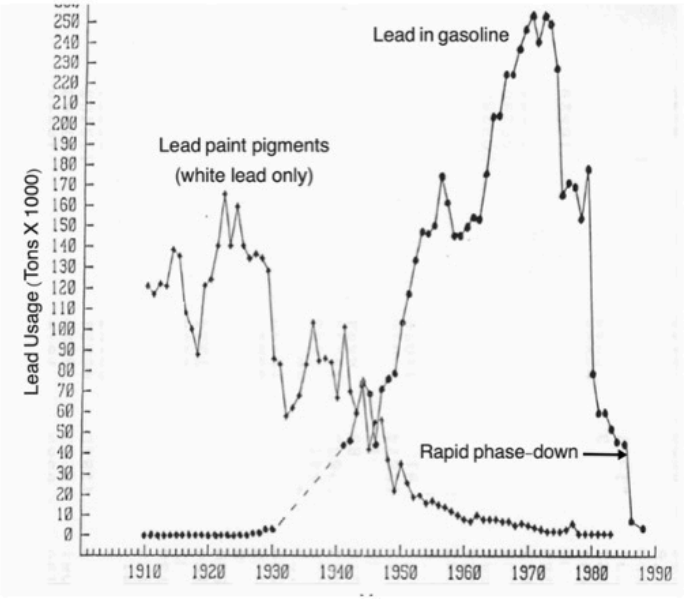
In [ ]:

In [10]:
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Image(filename='Historical_lead_usage.jpg',width=700)
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Out[10]:

Lead in Air, Soil, and Blood: Pb Poisoning in a Changing World

by Howard W. Mielke [1,*], Christopher R. Gonzales [2], Eric T. Powell [2] and Sara Perl Egendorf [3]



In [ ]:

Luckily lead levels in kids is decreasing.

This is a great public health success. BUT exposures are still present and levels should be dropping quicker. The next figure shows data from Blood Lead Levels in U.S. Children Ages 1–11 Years, 1976–2016 Authors: Kathryn B. Egan, Cheryl R. Cornwell, Joseph G. Courtney, and Adrienne S. EttingerAuthors Info & Affiliations Publication: Environmental Health Perspectives Volume 129, Issue 3 CID: 037003 https://doi.org/10.1289/EHP7932

In [2]:
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Image(filename='egan-fig-1.png',width=600)
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Out[2]:

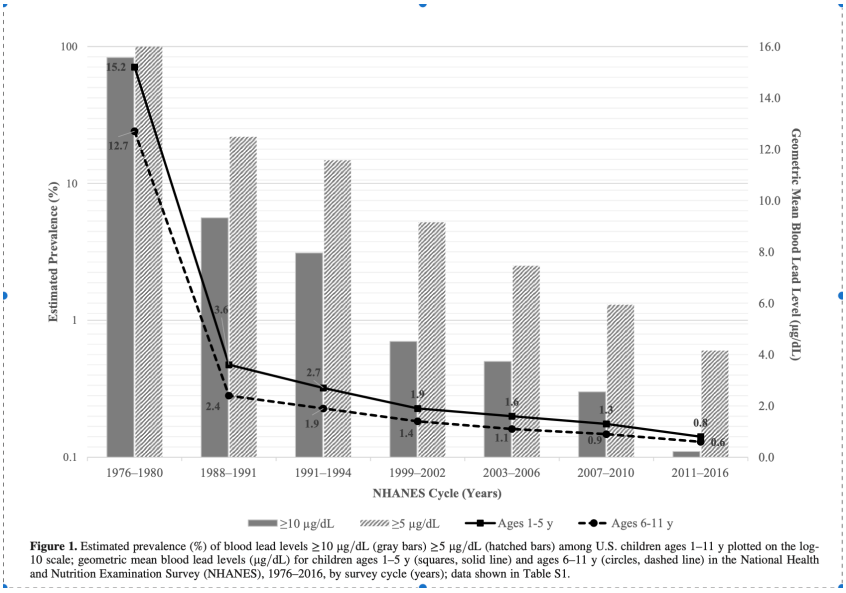


Figure 1. Estimated prevalence (%) of blood lead levels ≥10 µg/dL (gray bars) ≥5 µg/dL (hatched bars) among U.S. children ages 1–11 y plotted on the log-10 scale; geometric mean blood lead levels (µg/dL) for children ages 1–5 y (squares, solid line) and ages 6–11 y (circles, dashed line) in the National Health and Nutrition Examination Survey (NHANES), 1976–2016, by survey cycle (years); data shown in Table S1.

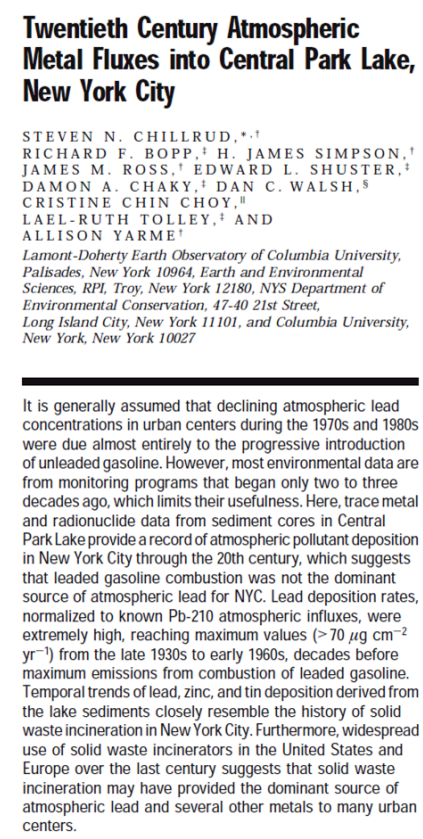
What is the source of Lead in NYC???

In most areas of the United States it is assummed that leaded gasoline was a major cause of lead pollution before it was banned in the 1970's. However New York City may be a unique case and could be different. Colleagues at Lamont put out this nice article saying that incineration of municipal waste was the main cause of lead to the atmosphere in New York City and not Leaded Gasoline during the 20th century. This was based on data collected from a sediment core. Here is a link to the paper and the pdf. See the abstract below. This

notebook is for information and you only need to read it. I have now also made a podcast talking to Steve about the paper.
https://itunes.apple.com/us/podcast/chillrud-lead/id1418618791?i=1000416725089&mt=2

In [9]:
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Image(filename='Chillrud-Abstract.png',width=300)
```

Out[9]:



Twentieth Century Atmospheric Metal Fluxes into Central Park Lake, New York City

STEVEN N. CHILLRUD,*·† RICHARD F. BOPP,‡ H. JAMES SIMPSON,† JAMES M. ROSS,† EDWARD L. SHUSTER,‡ DAMON A. CHAKY,‡ DAN C. WALSH,§ CRISTINE CHIN CHOY,∥ LAEL-RUTH TOLLEY,‡ AND ALLISON YARME†

Lamont-Doherty Earth Observatory of Columbia University, Palisades, New York 10964, Earth and Environmental Sciences, RPI, Troy, New York 12180, NYS Department of Environmental Conservation, 47-40 21st Street, Long Island City, New York 11101, and Columbia University, New York, New York 10027

It is generally assumed that declining atmospheric lead concentrations in urban centers during the 1970s and 1980s were due almost entirely to the progressive introduction of unleaded gasoline. However, most environmental data are from monitoring programs that began only two to three decades ago, which limits their usefulness. Here, trace metal and radionuclide data from sediment cores in Central Park Lake provide a record of atmospheric pollutant deposition in New York City through the 20th century, which suggests that leaded gasoline combustion was not the dominant source of atmospheric lead for NYC. Lead deposition rates, normalized to known Pb-210 atmospheric influxes, were extremely high, reaching maximum values (> 70 μg cm^{-2} yr^{-1}) from the late 1930s to early 1960s, decades before maximum emissions from combustion of leaded gasoline. Temporal trends of lead, zinc, and tin deposition derived from the lake sediments closely resemble the history of solid waste incineration in New York City. Furthermore, widespread use of solid waste incinerators in the United States and Europe over the last century suggests that solid waste incineration may have provided the dominant source of atmospheric lead and several other metals to many urban centers.

Smog Pictures from the NY Times.

In [7]:
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Image(filename='NYC-Smog-23UP-Pollution-ss-slide-2827-superJumbo.jpg',width=600)
```

Out[7]:



Here is an instagram video about [NYC trash chutes and inceration](https://www.instagram.com/reel/DdRdjomBI9J/_

In [ ]:

Here is a picture of the authors collecting the core in central park!

In [5]:
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Image(filename='central_park_core.jpg',width=600)
```

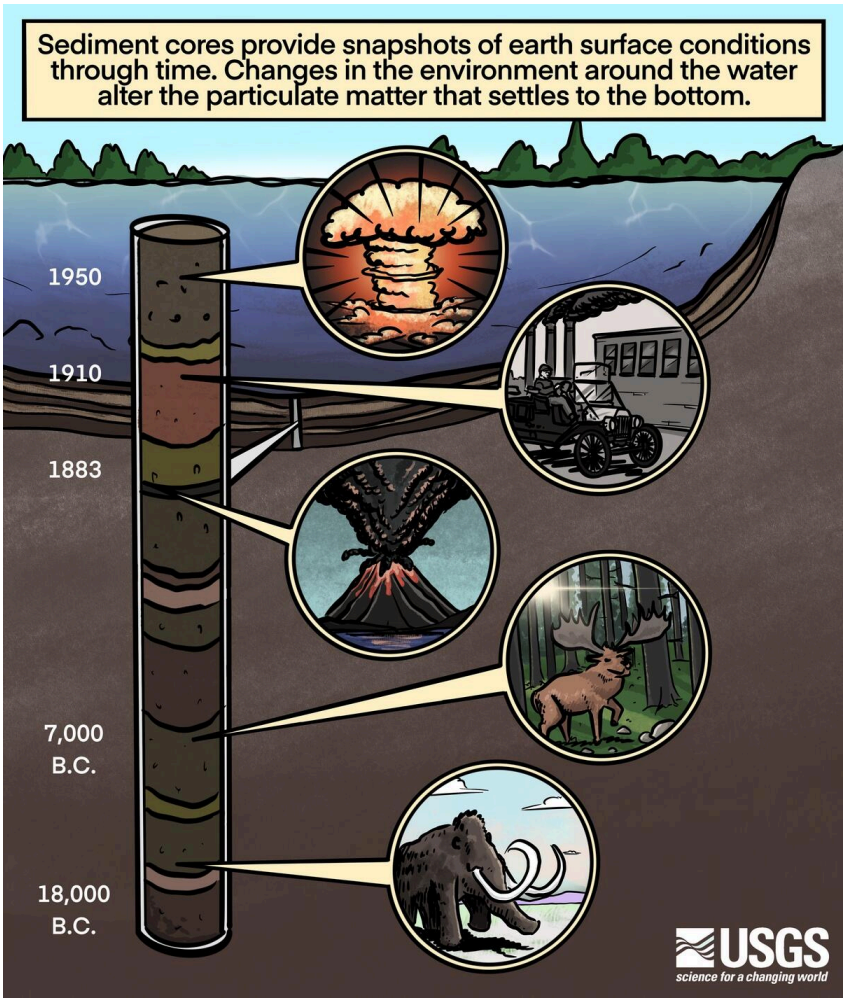
Out[5]:



How Sediment Cores Record Historical Records

In [3]:
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Image(filename='USGS-sedimentcore.jpg',width=600)
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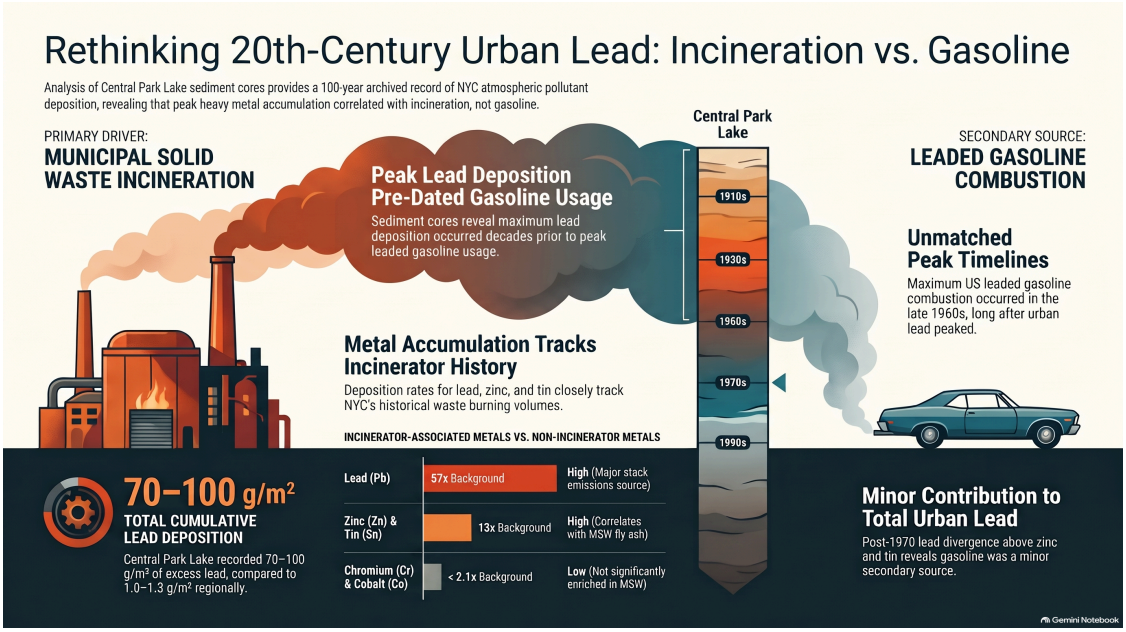
Out[3]:



AI Summary of paper (Not bad)

In [16]:
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Image(filename='Urban_Lead__Incineration_vs._Gasoline.png',width=800)
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Out[16]:



Look at Chillrud paper

Then

We want to think about the results from Chillrud et al., and use them to develop questions and hypotheses about other sediment cores.

We collected a sediment core from a Backyard in Greenpoint Brooklyn.

Your goal is to

1. read the Chillrud paper.
2. Listen to the podcast
3. Determine their main conclusions and hypothesis.
4. Develop your own hypothesis to test on the Brooklyn Core
5. Test that hypothesis with the sediment data from the Brooklyn Core.
6. Report your findings (This is going to include multiple graphs) and writing in markdown

This is a picture of a core geting collected in a Brooklyn Back Yard

In [3]:
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Image(filename='core-data/BrooklynCore.jpg',width=400)
```

Out[3]:



Here is what a core looks like once it is collected

In [4]:
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Image(filename='core-data/core-cropped.jpg',width=300)
```

Out[4]:

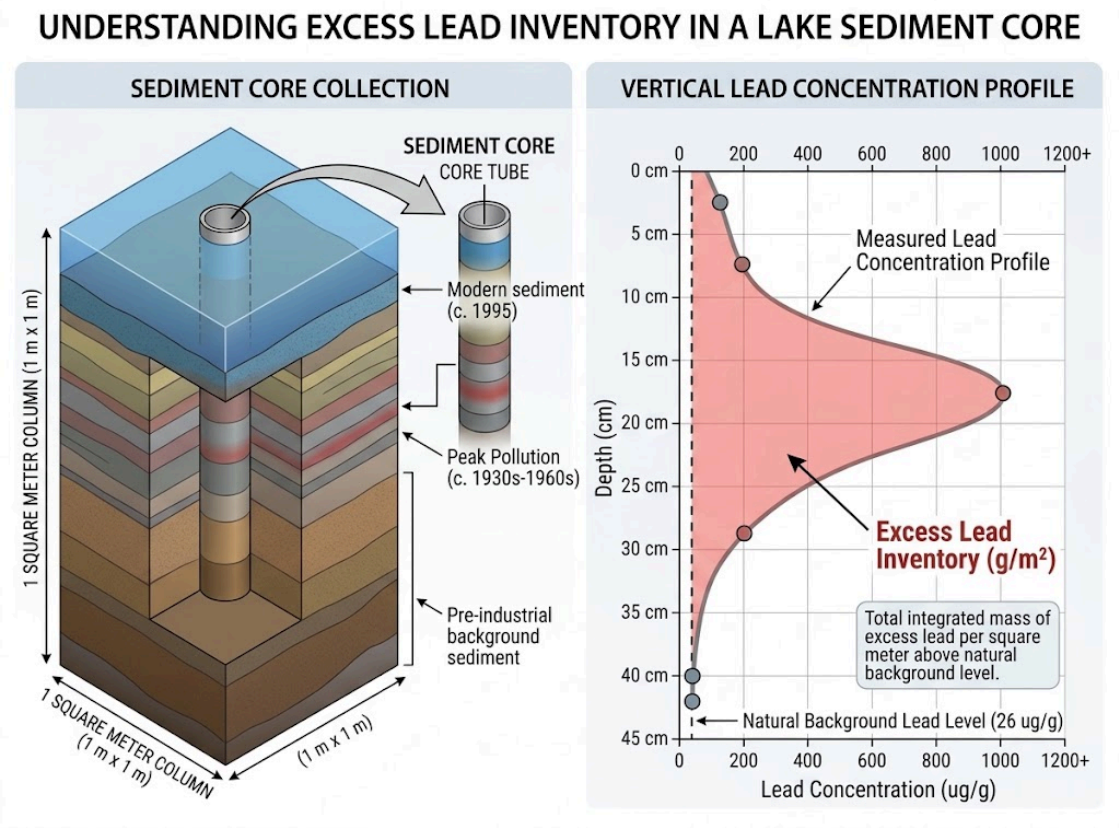


What is the inventory?

It is the total amount of lead in your core above background levels. It is normalized to the area and has units of g/m^2. No anthropenic lead added to sediment would equal 0. Here is my AI schematic.

In [13]:
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Image(filename='Invnetory_Schematic.jpeg',width=800)
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Out[13]:



How do I calculate?

- For each layer of the core
- You multiply the weight of the sediment by the concentration of lead. In Steve's paper they subtract a background concentration of 26 ug/g from the concentrations at first which is the same as mg/Kg as this is the natural levels in soils. This is stated in figure caption 3.
- Lead is in units of mg/Kg which is mg of lead per kg of soil
- the soil weight is in grams so you convert to kg.
- this gives you the weight of the lead in each section in units mg.
- You sum of the weights
- You convert to grams
- you divide by the cross-sectional area of the core in units of m^2
- the core radius is 1.27 cm

As math this looks like (watch the units!)

$$Inventory = \frac{1}{CoreArea} * \sum_{n=1}^{Sections} SedMass * (ConcentrationPb - 26)$$

$$CoreArea = \frac{1.27^2 * \pi}{100 * 100} = 0.000507$$

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Lead Concentrations versus Lead Isotopes

In order to understand the data you need to understand a little bit about lead isotopes. We are using total lead and lead isotopes in two ways in this study and in this paper.

- Lead Concentrations. Total amount of Lead. this controls the health effects.
- Lead Isotopes. These are the radionuclides of lead that can be used for dataing sediment.

Lead-210 is a natural radionuclide of lead and has a half life of 22 years. Lead is deposited at basically a constant rate from the atmosphere. But once sediment is buried it can't get anymore Lead-210. So the lead-210 decays and after 22 years you will have half as

much as when you started. So by measuring lead-210 you can get an idea of the time since depositions! That is how Steve has dates on the core. The lead-210 used for dating is also called Excess Lead-210.

But at the same time there is total Lead in the atmosphere. This is all the stable and unstable isotopes of lead. The amount of this total lead has changed in the atmosphere over time and we are interested in its temporal patterns and its controls. So Lead-210 is helping with dating the sediment and total lead is telling us about atmospheric contamination.

Also, within these cores there is deposition of Cs-137 from nuclear fallout. Also, there are other elements and metals that may have a contamination source while many elements are just natural. Can we figure out which are natural and which correlate with the different sources of contamination?

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Your Assignment

- Develop a hypothesis
- Test it on your core
- Report your findings
- I will show you a few tricks but I am giving you no structure
- you will hand in one Notebook that has your hypothesis, results, interpretation etc.
- Use what you know
- If you use AI you need to state if you used it and supply the prompt you used.
- If you go back and forth with AI a lot you need to supply your original prompt and then have AI generate a prompt. Then you need to re-run that prompt and make sure it gives you the same answer

In [ ]:

new page

Exit Ticket

Name:

uni:

Q1. Write a goal(s) you want to accomplish on this project before the next class.